Q.1 $\frac{{1 + d(x,y)}}{{1 - d(x,y)}} = 2 \Rightarrow d(x,y) =$

A. 1/3

B. 2/3

C. -1/3

D. 0

Q.2 In ${R^3}$, an open neighborhood is an .....

A. Open disc

B. Open interval

C. Closed ball

D. closed Interval

Q.3 Unitary space is

A. space of all types mentioned.

B. Banach space.

C. Vector space.

D. Normed space.

Q.4 A sequence $\left({{x_n}} \right)$ in a metric space X is said to be a..... if for every $\varepsilon > 0$, there is an N= $N(\varepsilon)$ such that for all m,n > N $d({x_m},{x_n}) < \varepsilon$

A. Arithmatic Sequence

B. Cauchy sequence

C. Harmonic Sequence

D. Geometric Sequence

Q.5 Vector space X is called real vector space if

A. $K = { C}$

B. $K = { N}$

C. $K = { Q}$

D. $K = { R}$

Q.6 T is bijective and norm preserving then it means that T is a\an.....between two norm spaces X and Y.

A. injective mapping

B. homomorphism

C. Isometric Isomorphism

D. surjective mapping

Q.7 The metric space defined on ${\Bbb Q}$ the set of rational numbers with usual metric

A. is incomplete.

B. has convergent Cauchy sequence.

C. is complete.

D. has only positive integers.

Q.8 d(x, y) =|x-y| is a usual metric defined on

A. ${R^3}$

B. ${R^n}$

C. R

D. ${R^2}$

Q.9 Which of the following statements is true about a metric space that is not complete?

A. Every bounded sequence in the space is convergent.

B. There exists a cauchy sequence in the space that does not converge in the space.

C. Every convergent sequence in the space is cauchy.

D. Every cauchy sequence in the space is convergent.

Q.10 Vector space X is called complex vector space if

A. $K = { Q}$

B. $K = { R}$

C. $K = { C}$

D. $K = { N}$

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